

The Cost of Consolidation

*Hub Monopolies, Fare Convergence, and the Collapse of Competition in U.S.
Domestic Aviation*

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Data: BTS DB1B, Marketing Carrier On-Time Performance (2024)

Executive Summary

Since 2024, Spirit Airlines has filed for bankruptcy twice, and the company has been rumored to face impending liquidation in the coming days or weeks. In a market that deregulation was supposed to make competitive, the airline that millions could depend on for budget travel is struggling to survive daily. This paper argues that Spirit's collapse was the predictable result of a U.S. domestic aviation market that is far less competitive than assumed by the public.

Using 2024 data from the Bureau of Transportation Statistics Origin and Destination Survey and On-Time Performance records covering 156 commercial airports across the United States, this analysis finds that 60.5% of served domestic routes in the United States operate under monopolistic conditions: a single parent carrier, with virtually zero competitive presence. Only 15.7% of served routes have three or more independent carriers, classifying as a "competitive" route. To add another angle to view the competition, the Herfindahl-Hirschman Index (HHI) was implemented. With a median score of 10,000 on observed routes, complete single-carrier control with zero competitive presence is widespread across the country.

The analysis identifies two dominant hub-based monopolistic territories: American Airlines via Charlotte Douglas International Airport (CLT) and Dallas/Fort Worth International Airport (DFW), Delta Air Lines via Atlanta Hartsfield-Jackson International Airport (ATL), and a third, near-monopolistic region established by United Airlines via Newark Liberty International Airport (EWR). These carriers have established zones of pricing power with minimal competition.

Most significant of these findings, 41 U.S. cities face 100% carrier concentration on routes to both CLT and ATL simultaneously, implying that travelers in these markets have no competitive alternatives in either direction. In these doubly captive markets, the "monopoly tax" is unforgiving; not only do average fares range from \$300 to \$500, but the absolute budget price floor (the 10th-percentile fare) routinely pushes beyond \$200 for short regional flights, completely pricing out budget-conscious consumers.

Findings suggest that Spirit's possible liquidation won't just remove another budget airline, it will remove, in many areas, the only line of defense against monopoly pricing in dozens of U.S. markets.

1. Introduction

As an incoming college graduate in the coming weeks, I used Spirit Airlines to get to places I would otherwise not be able to afford comfortably. A \$68 roundtrip ticket from Boston to Baltimore made a weekend trip possible that a \$160 ticket from Delta or Southwest would've made rather unappealing. For millions of American travelers, ranging from college students, working families, and people visiting family across the country on short notices, Spirit and its ultra-low-cost carrier (ULCC) companions represented something the U.S. airline industry hadn't generally offered at this scale: genuinely cheap Point A to Point B air travel.

Then, Spirit would file for bankruptcy twice in the span of 18 months, with the first filing coming in November 2024. The second following in the next year. An airline with dozens of destinations and millions of loyal travelers suddenly ceased to be viable in the present day. Stories that have emerged in aviation media focused on Spirit's operational failures, failed mergers with JetBlue and Frontier, and rising fuel costs attributable to the ongoing conflicts in the Middle East. But all of these missed perhaps a deeper question: why was Spirit in this position where cheap fares were no longer viable to begin with?

The answer which this paper argues lies in the overall structure of the domestic aviation market itself. Spirit didn't go bankrupt because the demand simply vanished into thin air, but rather because it could no longer maintain the price gap between mainline airlines, which justified the existence. This gap greatly narrowed when post-COVID fare spikes pushed ULCC fares closer towards mainline levels.

To understand what this means for travelers, I analyzed 2024 Bureau of Transportation Statistics (BTS) data covering passenger demand, fares, and carrier presence across 156 U.S. commercial airports, ranging from mid-size regional airports to international giants that see tens of millions of passengers pass through them annually. What the data revealed, is that the market Spirit was supposed to disrupt, was in many places, not as competitive as previously thought.

Three major findings are presented in this paper. Firstly, nearly two-thirds of served U.S. domestic routes observed in the data operate under monopolistic conditions. Second, three American legacy carriers: American, Delta, and United – have established geographical hub territories across the country, with greater pricing power and minimal competitive overlap. Third, 41 U.S. cities face complete carrier monopolies on both dominant Southeast hubs simultaneously, leaving passengers without any competitive alternatives.

2. Background

2.1 Deregulation and Consolidation

The Airline Deregulation Act of 1978 removed governmental control over fares and domestic routes, creating the conditions for competitive pricing that have been present for the past half century. The following decades saw the emergence of lower-cost carriers, most prominently Southwest Airlines, which demonstrated that budget air travel was not only popular, but also profitable.

The turn of the 21st century brought a volatile period of consolidation that reshaped the competitive landscape. Delta merged with Northwest in 2010, United merged with Continental in 2012, and American merged with US Airways in 2015. By the mid-2010s, four carriers: American, Delta, United, and Southwest, controlled nearly 80% of American domestic capacity, effectively ending the era of multi-carrier competition on most routes.

2.2 The Rise of the ULCC Model

In the new era of consolidating markets, a class of budget carriers emerged to offer alternatives to passengers in a rapidly changing landscape: Spirit, Frontier, and Allegiant.

These airlines sought after a completely different model: stripping fares down to the bare minimum, and charging separately for everything else one can imagine, even down to the carry-ons which were taken for granted on mainline carriers. The base fare itself **was** genuinely cheap, whereas the fees that were attached took on the form of the business model itself.

The model worked when the price gap between ULCC and mainline fares was wide enough to justify the tradeoffs of the model. Passengers could look past the uncomfortable seats, fees for carry-ons, and the seemingly unreliable operations, because the effects on their wallets weren't nearly as damaging. In 2019, Spirit averaged \$153 per domestic ticket, whereas Delta averaged \$424. A \$271 gap served as a lifeline for keeping the ULCC model workable.

2.3 COVID and Convergence

The COVID-19 pandemic disrupted this model, in ways that have proved to be permanent. When demand for travel collapsed in 2020, all carriers dramatically dropped their fares to fill their planes. I vividly remember boarding a Delta flight from Boston to Fort Lauderdale in February 2021, which had costed \$60 roundtrip – a price that only seemed possible on Spirit.

When demand returned in 2022, it returned faster than capacity could respond to, and fares spiked across the board. Spirit's 2022 average fare hit \$222. Delta's hit \$316. The gap between the two carriers had compressed by 65%, all the way down to \$94; a critical blow to the ULCC model.

By the time Spirit had filed for bankruptcy in 2024, the airline which prided itself on being significantly cheaper than the competition, had, in several markets, become nearly as expensive. Without the price gap, there were simply no compelling reasons to accept the many tradeoffs of Spirit's business model.

3. Data and Methodology

3.1 Data Sources

This analysis draws from two primary data sources from the Bureau of Transportation Statistics (BTS).

The BTS Airline Origin and Destination Survey (DB1B) is a 10% sample of all domestic airline tickets, updated quarterly. For this study, all four quarters of 2024 were observed, covering connecting passenger volumes and average fares across domestic routes. Estimates for passengers were scaled by a factor of ten, to approximate the true annual volumes consistent with the sampling methodology.

The BTS Marketing Carrier On-Time Performance dataset provides flight-level data for all U.S. certified carriers. The full year 2024 was observed to determine which carriers operate each route, and with what frequency.

Airport geographic data was sourced from OpenFlights' airport database. City population data was drawn from the SimpleMaps U.S. Cities dataset, derived from U.S. Census Bureau estimates.

3.2 Airport Selection

The analysis was restricted to U.S. commercial airports with scheduled services and a valid IATA code, being further filtered to those with a minimum of 5,000 quarterly passengers in the DB1B sample, yielding a working sample of 156 commercial airports. This threshold was applied to exclude remote airports in regions such as Alaska, and other locations that are not commercially viable route targets.

In the section regarding doubly captive cities, to be classified as such in this analysis, legs to ATL and CLT must exceed 100 miles and have 5,000 connecting passengers in both cases.

3.3 Competition Classification

Carrier presence was determined from the On-Time Performance dataset. Importantly, regional carrier subsidiaries were linked to their parent companies to avoid overstating competition. Carriers coded as MQ (Envoy Air), and OH (PSA Airlines), were branched under the American Airlines parent. Carriers coded as 9E (Endeavor Air) were branched under Delta.

In some cases, such as for OO (Skywest Airlines), they were mapped to their corresponding mainline partners and were assigned to the right mainline carrier based on each individual route based on the BTS data, and not a general assumption.

Routes themselves were classified into three tiers of competition: Monopoly (one parent carrier, 0% competitive presence on the route), Duopoly (two independent parent carriers), and Competitive (three or more independent parent carriers). The monopoly classification was scanned against the On-Time Performance data to verify 100% carrier share on identified routes.

This analysis excludes theoretical airport pairings with no recorded flights, focusing entirely on the 5,525 served domestic routes that maintain regular services. Intra-Alaska and intra-Hawaii routes were excluded from the analysis as their competitive dynamics reflect geographic isolation rather than hub-based market structure.

To account for multi-airport markets, routes were flagged where either the origin or destination airport belongs to a catchment area served by multiple distinct airports. Routes within these corridors are noted as having competitive alternatives that may not be captured by airport-pair competition classification by itself.

To see a full list of catchment zones, see Appendix C.

Market concentration was additionally measured using the Herfindahl-Hirshman Index (HHI), a standard metric calculated as the sum of squared carrier market shares on each unique route. An HHI score of 0 indicates perfect competition, whereas a score of 10,000 indicates pure monopoly; one carrier controlling 100% of flights. Every case study route presented in this paper scores 10,000 on the HHI, confirming complete carrier control over these routes.

3.4 Route Viability Index

A Route Viability Index (RVI) was constructed to rank route-level opportunities, incorporating origin and destination city population, connecting passenger demand, route distance, and average fares relative to distance. The RVI was used to prioritize routes for case study analysis and will be presented in full in the appendix. The primary framework of

this paper relies on connecting passenger volume and concentration of carriers, rather than the composite RVI score.

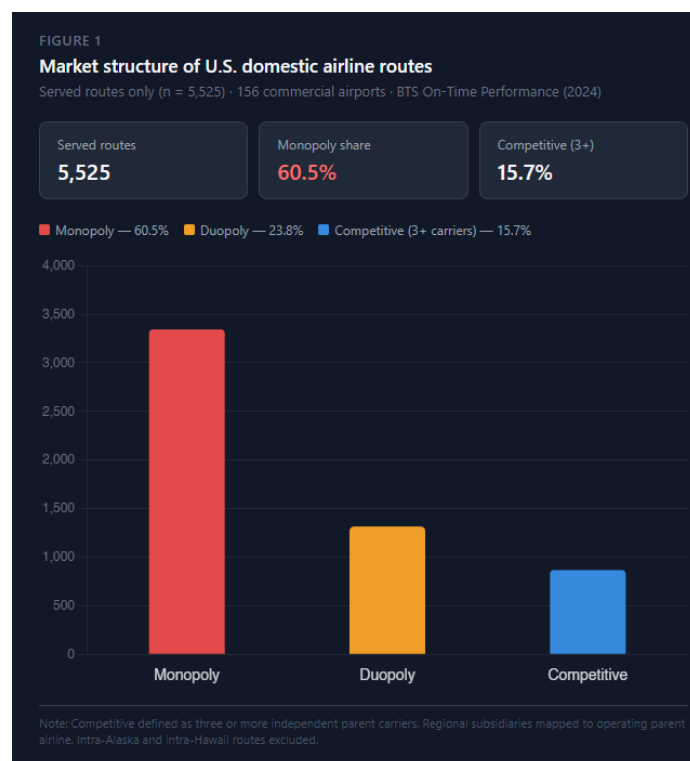
4. Finding 1 – Market Structure

Of the 5,525 domestic routes with confirmed carrier services among the 156 airports in this analysis, 60.5% operate under monopoly conditions; a single carrier with zero competitive presence. An additional 23.8% operate as duopolies. Only 15.7% of observed routes have three or more independent carriers.

These numbers represent a more concentrated market than perceived by the public. The consolidation era of 2010-2015 didn't merely reduce the number of carriers – it produced a market in which the majority of routes are served by one airline without any competitive constraints.

It is important to note that monopoly routes are not uniformly distributed. As the route characteristic analysis in Section 5 demonstrates, many monopolistic routes serve smaller markets, where demand doesn't justify the presence of multiple competing carriers. The analytically significant finding is not the aggregate percentage of monopolies, but rather the subset of high-demand monopoly routes where competitive service would be economically viable but are nonexistent in the data.

Figure 1 – Market Structure of U.S. Domestic Airline Routes



4.1 The Monopoly Tax

The aggregate monopoly percentage does not fully state the impact to consumers regarding market concentration, as average fares do not account for route distance. A monopoly route of 175 miles and a competitive route of 1,500 might carry similar average fares, but the per-mile cost paints a different picture.

Controlling for distance, monopoly routes consistently charge more per mile than competitive routes across every distance band. On short-haul routes, under 300 miles; the segment most relevant to regional, hub-feeding markets, the median cost per mile on monopoly routes is \$1.33, compared to \$0.87 on equidistant competitive routes. That's a 54% premium that passengers on these routes must absorb without any competitors to offer another option.

The premium does compress at longer distances; falling to 21% on routes between 300 and 600 miles, and to roughly 12% on routes between 600 and 1,000 miles. This reflects the reality of the market: longer routes usually attract more carriers, thus facing more competitive pressure. The short, regional routes, precisely the hub-feeder markets where American, United and Delta hold monopolistic control, face the largest costs-per-mile and the narrowest alternatives.

This point is illustrated with Oklahoma City to Dallas/Fort Worth. At 175 miles, squarely under the under-300-mile band, passengers pay American an average of \$413, equating to \$2.35 per mile, contrasting the \$0.87 competitive benchmark in an extreme manner. The monopoly tax on this route amounts to around \$260 per passenger **above** what a competitive market would be expected to produce.

However, it is important to take into consideration that average fares are often skewed by last-minute purchases, at elevated costs. As such, for each average fare listed, the 10th percentile of each fare is also recorded, as the “budget price floor”. This data point represents the cheapest entry point for budget travelers. In monopolistic markets, this floor has dissipated, with the 10th percentile for many routes sitting above \$200.

5. Finding 2 – Hub Monopoly Territories

The distribution of monopoly routes is not random. It reflects the deliberate network structure of the three remaining legacy carriers, which each have established territories of hub-based pricing power.

5.1 American Airlines via CLT and DFW

American Airlines operates as the largest hub at Charlotte Douglas International Airport (CLT). The data reveals a consistent pattern of complete carrier monopoly across routes that connect CLT to secondary markets in the Southeast, Mid-Atlantic, and Northeast.

The most egregious example of monopoly pricing by American occurs on an intra-state route from Raleigh-Durham International Airport (RDU) to CLT. The short, 130-mile hop between the two cities is purely in control of American and runs an average fare of \$349. 329,850 passengers who connect between these two cities are subject to extremely high cost-per-mile journeys.

Other notable examples include Richmond to Charlotte (232,960 annual connecting passengers, \$405 average fare, purely in control of American), Jacksonville to Charlotte (188,670 passengers, \$420 average fare), and Pittsburgh to Charlotte (161,170 passengers, \$416 average fare).

American's second major monopoly territory emerges from Dallas/Fort Worth (DFW). Oklahoma City to DFW presents itself as the most striking case: 186,300 annual connecting passengers, who travel a 175-mile route that takes just over an hour, yet pay an average fare of \$413. All OKC-DFW flights are operated by American or its partners and subsidiaries. To add context, the average fare on competitive routes which cover a similar distance, such as Boston to New York, is significantly lower, often registering under \$200.

All routes in this analysis scored a value of 10,000 on the Herfindahl–Hirschman Index (HHI), a measurement tool used to determine the competitiveness of markets. A score of zero implies perfect competition, a score below 1,500 is “competitive”, while scores above 2,500 are causes for concern. The maximum score – 10,000, implies a perfect monopoly within the market, which is achieved by over half of the observed routes in the analysis.

5.2 Delta Air Lines via ATL

Delta Air Lines dominates the world's busiest airport: Atlanta Hartsfield-Jackson International Airport, and its many routes connecting it to the Deep South and Southeast. Birmingham to Atlanta represents the highest-demand monopoly route from ATL: 244,900 annual connecting passengers, traveling only 134 miles at an average fare of \$359, all of which is under the sole control of Delta. Savannah to Atlanta (205,630 passengers, \$345), and Pensacola to Atlanta (212,090 passengers, \$336) both follow the same pattern.

The geographic location of Delta's monopoly of ATL is reinforced by its proximity to American's CLT hub. The two airports are roughly 245 miles apart, creating a zone of

duopolistic control across much of the Southeast, in which most secondary cities face monopoly pricing, regardless of which major hub they choose to route through.

5.3 United Airlines via EWR

While less geographically concentrated than the patterns showed by American or Delta, United demonstrates significant monopoly control on certain long-haul routes from Newark Liberty International Airport (EWR). The most notable example is Newark to Denver: 215,230 annual connecting passengers, a 1,600-mile route, 100% operated by United, with an average fare of \$362. With 3,103 observed flights in 2024 and no competitive presence, this route illustrates that hub monopolies aren't just limited to short-haul regional markets.

However, an important caveat is that EWR is one of three major airports serving the New York City area, alongside John F. Kennedy International Airport (JFK), and LaGuardia (LGA). A passenger who originates in Newark will have access to these other alternatives at JFK and LGA that would otherwise not exist for passengers from Birmingham or Oklahoma City. This is reflected in the data: EWR-DEN is classified as a monopoly on a point-to-point basis, but the broader New York market offers United genuine competition from multiple carriers in the corridor to Denver.

This analysis flags EWR and similar multi-airport zones using a co-terminal market classification. Routes where the origin or destination belong to a multi-airport zone; such as New York, Washington D.C., Chicago, and Houston, carry implicit competitive alternatives that would otherwise not be present in single-airport markets. This EWR finding is presented as evidence that hub monopolies extend to longer-haul routes, while acknowledging that the practical impact to budget-conscious travelers is less severe than findings in the Southeast, without co-terminal alternatives.

5.4 The Southwest Effect via GSP

Airlines often cite high regional fares on the steepening costs of operating small, 50-seat regional jets. But, if specific markets are isolated, market concentration, rather than operating costs, is revealed as the driver of these higher prices. Greenville-Spartanburg (GSP) serves as a real-world example of the monopoly tax in action.

GSP is a regional spoke that feeds directly into two major hubs: American Airlines at CLT, and Delta Air Lines at ATL. Even though local economics are similar, the competitive arrangements on these two routes from GSP create polarizing realities for budget travelers.

The 75-mile jump from GSP to CLT is a pure monopoly, controlled by American. Because passengers lack any alternatives, as far as air travel is concerned, the 10th-percentile fare, indicative of the “cheapest” tickets sold for this route, is an alarmingly high \$265.

On the other side of the coin, the route from GSP to ATL tells a completely different story. The flight itself is 150 miles, or double that of GSP-CLT. But this route is not completely monopolized by Delta Air Lines; Southwest Airlines has contested this route and acts as a direct competitor. Just having a single low-cost carrier drops the 10th-percentile floor all the way down to \$85.

Even though the Atlanta flight is double the distance, the lowest fares commonly available are \$180 cheaper. That \$180 is the monopoly tax in action. This example proves that, without a low-cost carrier to keep a legacy carrier honest about their pricing, legacy carriers on their captive spokes often choose to base their pricing around their market control, regardless of their flight distance.

6. Finding 3 – The Doubly Captive Cities

American’s CLT network and Delta’s ATL network converge to produce a constricting outcome for 41 U.S. cities: complete carrier monopolies on routes to both hubs simultaneously. These cities, which this analysis classifies as “doubly captive”, have no independent competitive option in either direction.

Verification against On-Time Performance data has confirmed that 100% of flights on all routes from these 41 cities to both CLT and ATL are operated solely by the respective dominating carriers and their subsidiaries, with no independent competitors offering even one flight. It is complete market control for these routes.

The doubly captive cities span across the United States, from Birmingham, Alabama, to Sacramento, California. Average fares to both hubs range from \$300 to \$500, with no reliable alternatives offered by lower-cost carriers.

To see a full list of the 41 doubly captive cities, see Appendix H.

7. The ULCC Connection

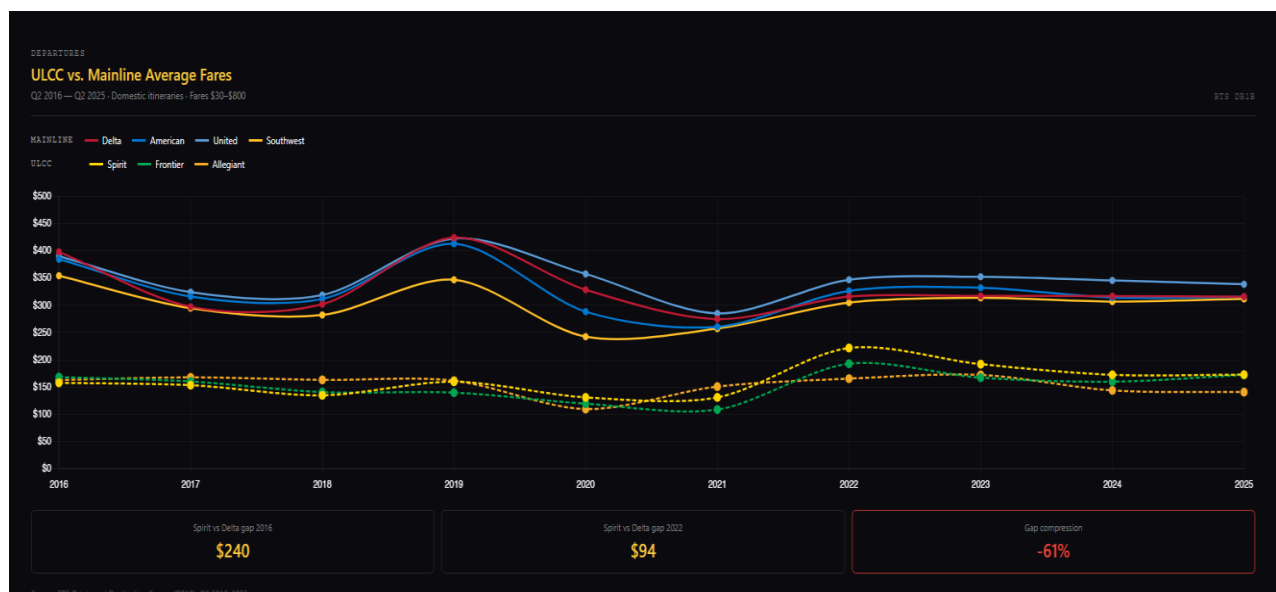
Spirit Airlines’ mission was to make air travel more accessible to people who could otherwise not afford it comfortably. This mission created an implicit economic logic: Spirit contested markets where fares were high and competition was low, drove prices down, and made air travel possible for passengers who would have otherwise been crowded out.

The routes where Spirit competed most aggressively overlap with the high-fare monopolistic markets identified in this analysis. Where Spirit flew, its presence created pressure on mainline airlines and forced them to adjust their prices accordingly to remain the best universal option. Its bankruptcy removed that constraint entirely on the affected routes.

The ULCC fare convergence analysis that motivated this paper shows average Spirit fares rising by \$90 over three years – from \$129 in 2019 to \$219 in 2022, before settling down to \$169 by 2025. During this period, Spirit simultaneously bled the price advantage that justified the operational tradeoffs, and absorbed cost increases that the tight margins with the ULCC model could not sustain.

The result is a market in which the ULCC competitive check has substantially weakened, at nearly the exact moment when the monopoly concentration documented in this paper is at its strongest. The cities most dependent on budget travel are, in many cases, the cities who now face new monopoly-pricing environments.

Figure 2 – ULCC vs Mainline Average Fares (2016-2025)



8. Implications

The findings of this analysis have implications at both the policy and industry level.

From purely a policy perspective, the concentration documented in this analysis raised questions about the current adequacy of the competitive framework in U.S. domestic aviation. In particular, the doubly captive city finding suggests that some markets warrant closer scrutiny of carrier behavior and pricing.

From an industry perspective, the data identifies certain markets where new entrant competition does appear to be economically viable. Routes that combine high amounts of connecting passengers, with elevated average fares, and single-carrier control – such as OKC-DFW, RIC-CLT, BHM-ATL, and EWR-DEN, represent potential openings for carriers who are willing to put a hand up and contest monopoly territory. The demand for these routes clearly exists, but the competitive supply is notably absent.

The question remains, however, whether any remaining ULCC carrier can fill in the gaps left by Spirit's diminishing territories. Frontier and Allegiant remain fully operational but have not demonstrated the route density or market presence that Spirit maintained at its peak. Without a competitive ULCC check, the monopoly pricing patterns documented in this analysis are likely to persist, if not intensify in the coming years.

8.1 Limitations of Analysis

While the data points to a highly concentrated market, this analysis must acknowledge some external factors that ticket data alone can't perfectly capture.

Intermodal Competition: On short-haul routes, an airline's biggest competitor might not always be another airline. In some cases, it might be a car, or a train. For example, in Appendix H, the 10th-percentile fare floor for Chattanooga (CHA) to Atlanta (ATL) sits at an unusually low \$36 and sticks out from the other doubly captive cities. Thousands of flights were reported for this route, implying that a data error isn't present. It is more likely that Delta understands that driving from Chattanooga to Atlanta is a simple two-hour venture down I-75. If Delta were to charge more monopoly premiums, more of the market may choose to simply drive. Similar pricing patterns appear in the Northeast corridor, where the presence of Amtrak keeps legacy airlines in check, without taking the higher presence of competition into account.

Aircraft Gauge and Operating Costs: Legacy airlines often justify higher fares on captive spokes by pointing to the steep unit costs of operating lower-capacity regional jets, a problem exacerbated by the regional pilot shortage of the 2020s. A ULCC like Spirit cannot easily justify deploying a 180-seat Airbus A320 on a short, thin route such as Oklahoma

City to Dallas. Legacy carriers utilize lower-capacity regional jets to provide the high-frequency schedules that hub networks require, whereas a ULCC, which often relies entirely on local, point-to-point traffic, would face an uphill battle filling a mainline aircraft on these routes.

Essential Air Service (EAS) Subsidies: The U.S. domestic network includes many small markets that are subsidized by the government through the EAS program. While setting a 5,000-passenger quarterly minimum for this analysis successfully filtered out many of these subsidized routes, government intervention can still play a background role in the economics of the aviation industry's smallest regional airports, which can distort the free-market metrics present.

Conclusion

Spirit Airlines has faced two bankruptcies because it stopped being “cheap” before withering into its current state. The price gap that gave the airline its value has eroded, with post-COVID fare spikes pushing ULCC fares towards mainline levels. But the deeper story present in the data revealed that the markets that Spirit was intended to contest, were structured in ways that made true competition difficult to achieve, regardless of which carriers entered them.

Over 60% of served U.S. domestic routes operate under monopolistic conditions. American, Delta, and United have each established their own hub territories with pricing power that extends across hundreds of city pairs, with several dozen cities facing complete carrier concentration on routes to both dominant Southeast hubs, lacking any viable alternatives in either direction. As the data demonstrates, without the competitive check of a low-cost carrier, legacy airlines are free to extract high monopoly taxes on their routes, frequently pushing the absolute cheapest tickets on these captive spokes well over \$200.

The travelers who are most affected by these troubling dynamics are not the business travelers who take up hub-to-hub corridors. They are college students like me, working families, and other budget-conscious travelers who relied on Spirit's low fares to make travel feasible. For those travelers, the degrading of the ULCC model is a meaningful reduction in access to air travel, that the currently existing competitive structure does little to address.

Appendix

A. Data Sources

BTS Airline Origin and Destination Survey (DB1B): transtats.bts.gov. Q1-Q4 2024. 10% sample of domestic airline tickets. Passenger estimates scaled by factor of ten to approximate true annual volumes.

BTS Marketing Carrier On-Time Performance: transtats.bts.gov. Full year 2024. All U.S. certificated carriers.

OpenFlights Airport Database: openflights.org/data. Airport IATA codes, coordinates, and scheduled service status.

SimpleMaps U.S. Cities Dataset: simplemaps.com/data/us-cities. City population estimates derived from U.S. Census Bureau data.

B. Regional Carrier to Parent Airline Mapping

MQ (Envoy Air) → American Airlines

OH (PSA Airlines) → American Airlines

OO (SkyWest Airlines) → Mapped to operating mainline partner per BTS data. SkyWest is an independent regional carrier contracted by American, Delta, United, and Alaska. No independent SkyWest-branded flights appeared for the year 2024.

9E (Endeavor Air) → Delta Air Lines

YX (Republic Airways) → United Airlines

G7 (GoJet Airlines) → United Airlines

C. Co-Terminal Multi-Airport Markets

The following metro areas were designated as multi-airport markets. Routes where the origin or destination belongs to one of the listed catchment areas are flagged as such.

New York: EWR, JFK, LGA, HPN

Washington D.C.: IAD, DCA, BWI

Chicago: ORD, MDW

Los Angeles: LAX, BUR, SNA, ONT, LGB

San Francisco: SFO, OAK, SJC

Dallas: DFW, DAL

Houston: IAH, HOU

Miami: MIA, FLL, PBI

D. Route Viability Index Methodology

The Route Viability Index (RVI) was constructed using the following weighted formula applied to standardized (z-scored) log-transformed variables:

$$RVI = 0.30 \times z(\text{origin population}) + 0.30 \times z(\text{destination population}) + 0.20 \times z(\text{connecting passengers}) - 0.20 \times z(\text{distance}) - 0.10 \times (\text{short-haul indicator}) - 0.10 \times (\text{long-haul indicator}) + 0.15 \times z(\text{cost per mile})$$

Short-haul routes were defined as those under 250 miles. Long-haul routes were defined as those over 2,500 miles. The cost-per-mile component rewards routes where existing fares are elevated relative to distance. All variables were logarithmically transformed prior to standardization. RVI was used to rank monopoly routes by opportunity but is not presented as the primary analytical framework of this paper.

E. Monopoly Tax by Distance Band

Median cost-per-mile for monopoly and competitive routes, controlling for route distance. Competitive defined as 3+ independent parent carriers. Source: BTS DB1B (2024), direct fares (\$30-\$800).

Distance band	Monopoly \$/mi	Competitive \$/mi	Premium
Under 300 miles	\$1.33	\$0.87	+54%
300–600 miles	\$0.58	\$0.48	+21%
600–1,000 miles	\$0.32	\$0.28	+12%
1,000–1,500 miles	\$0.22	\$0.22	+3%
Over 1,500 miles	\$0.17	\$0.15	+7%

F. Full Airport Pool (156 airports)

IATA	City	IATA	City	IATA	City
ABE	Allentown, PA	FLL	Fort Lauderdale, FL	ORD	Chicago, IL
ABQ	Albuquerque, NM	FSD	Sioux Falls, SD	ORF	Norfolk, VA
ACY	Atlantic City, NJ	FWA	Fort Wayne, IN	PAE	Everett, WA
AGS	Augusta, GA	GEG	Spokane, WA	PBI	West Palm Beach, FL

ALB	Albany, NY	GNV	Gainesville, FL	PDX	Portland, OR
AMA	Amarillo, TX	GRR	Grand Rapids, MI	PGD	Punta Gorda, FL
ANC	Anchorage, AK	GSO	Greensboro, NC	PHL	Philadelphia, PA
ATL	Atlanta, GA	GSP	Greenville, SC	PHX	Phoenix, AZ
ATW	Appleton, WI	HNL	Honolulu, HI	PIA	Peoria, IL
AUS	Austin, TX	HOU	Houston, TX	PIE	Pinellas Park, FL
AVL	Asheville, NC	HPN	White Plains, NY	PIT	Pittsburgh, PA
AZA	Mesa, AZ	HRL	Harlingen, TX	PNS	Pensacola, FL
BDL	Hartford, CT	HSV	Huntsville, AL	PSC	Pasco, WA
BHM	Birmingham, AL	HVN	New Haven, CT	PSP	Palm Springs, CA
BIL	Billings, MT	IAD	Dulles, VA	PVD	Providence, RI
BLI	Bellingham, WA	IAH	Houston, TX	PVU	Provo, UT
BNA	Nashville, TN	ICT	Wichita, KS	PWM	Portland, ME
BOI	Boise, ID	ILM	Wilmington, NC	RAP	Rapid City, SD
BOS	Boston, MA	IND	Indianapolis, IN	RDM	Redmond, OR
BTR	Baton Rouge, LA	ITO	Hilo, HI	RDU	Raleigh/Durham, NC
BTW	Burlington, VT	JAN	Jackson, MS	RIC	Richmond, VA
BUF	Buffalo, NY	JAX	Jacksonville, FL	RNO	Reno, NV
BUR	Burbank, CA	JFK	New York, NY	ROA	Roanoke, VA
BWI	Baltimore, MD	LAS	Las Vegas, NV	ROC	Rochester, NY
BZN	Bozeman, MT	LAX	Los Angeles, CA	RSW	Fort Myers, FL
CAE	Columbia, SC	LBB	Lubbock, TX	SAN	San Diego, CA
CAK	Akron, OH	LEX	Lexington, KY	SAT	San Antonio, TX
CHA	Chattanooga, TN	LGA	New York, NY	SAV	Savannah, GA
CHO	Charlottesville, VA	LGB	Long Beach, CA	SBA	Santa Barbara, CA
CHS	Charleston, SC	LIT	Little Rock, AR	SBN	South Bend, IN
CID	Cedar Rapids, IA	MAF	Midland, TX	SBP	San Luis Obispo, CA
CLE	Cleveland, OH	MCI	Kansas City, MO	SDF	Louisville, KY
CLT	Charlotte, NC	MCO	Orlando, FL	SEA	Seattle, WA

CMH	Columbus, OH	MDT	Harrisburg, PA	SFB	Orlando, FL
COS	Colorado Springs, CO	MDW	Chicago, IL	SFO	San Francisco, CA
CRP	Corpus Christi, TX	MEM	Memphis, TN	SGF	Springfield, MO
CVG	Cincinnati, KY	MFE	McAllen, TX	SHV	Shreveport, LA
DAL	Dallas, TX	MFR	Medford, OR	SJC	San Jose, CA
DAY	Dayton, OH	MHT	Manchester, NH	SLC	Salt Lake City, UT
DCA	Washington, DC	MIA	Miami, FL	SMF	Sacramento, CA
DEN	Denver, CO	MKE	Milwaukee, WI	SNA	Santa Ana, CA
DFW	Dallas-Fort Worth, TX	MLI	Moline, IL	SRQ	Sarasota, FL
DSM	Des Moines, IA	MSN	Madison, WI	STL	St. Louis, MO
DTW	Detroit, MI	MSO	Missoula, MT	STS	Santa Rosa, CA
ECP	Panama City Beach, FL	MSP	Minneapolis, MN	SYR	Syracuse, NY
ELP	El Paso, TX	MSY	New Orleans, LA	TLH	Tallahassee, FL
EUG	Eugene, OR	MYR	Myrtle Beach, SC	TPA	Tampa, FL
EWB	Newark, NJ	OAK	Oakland, CA	TUL	Tulsa, OK
FAI	Fairbanks, AK	OGG	Kahului, HI	TUS	Tucson, AZ
FAR	Fargo, ND	OKC	Oklahoma City, OK	TVC	Traverse City, MI
FAT	Fresno, CA	OMA	Omaha, NE	VPS	Valparaiso, FL
FCA	Kalispell, MT	ONT	Ontario, CA	XNA	Fayetteville, AR

G. Top 25 Underserved Monopoly Routes

Routes with carrier_count = 1, connecting passengers > 5,000, and distance > 125 miles, ranked by RVI. All passenger figures are estimated annual volumes scaled from the 10% DB1B sample.

Orig	Dest	Mi	Ann. pax	Avg fare	p10 fare	HHI	Carrier
RDU	CLT	130	329,850	\$349	\$262	10,000	American
BHM	ATL	134	244,900	\$359	\$188	10,000	Delta
RIC	CLT	256	232,960	\$405	\$242	10,000	American
CHS	ATL	259	222,120	\$358	\$198	10,000	Delta

EWB	DEN	1,601	215,230	\$362	\$181	10,000	United
PNS	ATL	272	212,090	\$336	\$198	10,000	Delta
SAV	ATL	214	205,630	\$345	\$188	10,000	Delta
PHX	CLT	1,770	202,600	\$416	\$187	10,000	American
MFE	DFW	470	192,910	\$389	\$181	10,000	American
DCA	DFW	1,190	188,790	\$402	\$181	10,000	American
JAX	CLT	329	188,670	\$420	\$242	10,000	American
OKC	DFW	175	186,300	\$413	\$260	10,000	American
PIT	CLT	367	161,170	\$416	\$216	10,000	American
TUL	DFW	237	166,640	\$282	\$141	10,000	American
ILM	CLT	184	155,510	\$395	\$254	10,000	American
HSV	ATL	151	140,470	\$334	\$198	10,000	Delta
ECP	ATL	240	138,090	\$339	\$178	10,000	Delta
MCI	MDW	404	138,440	\$209	\$73	10,000	Southwest
HOU	DAL	240	150,980	\$196	\$75	10,000	Southwest
GSO	ATL	306	149,620	\$364	\$197	10,000	Delta
SAT	IAH	190	125,490	\$251	\$119	10,000	Southwest
VPS	ATL	250	122,210	\$317	\$186	10,000	Delta
TLH	ATL	224	106,690	\$345	\$208	10,000	Delta
SGF	DFW	364	108,130	\$427	\$240	10,000	American
AUS	DAL	189	97,100	\$213	\$87	10,000	Southwest

H. Doubly Captive Cities – 41 Cities

100% carrier concentration to both CLT (American) and ATL (Delta). Both legs must exceed 100 miles and reach 5,000 annual connecting passengers. CHA-ATL p10 of \$36 reflects intermodal competition.

Code	City	ATL avg	ATL p10	ATL pax	CLT avg	CLT p10	CLT pax
BHM	Birmingham, AL	\$359	\$188	244,900	\$448	\$252	74,370
CHS	Charleston, SC	\$358	\$198	222,120	\$344	\$235	174,290

PNS	Pensacola, FL	\$336	\$198	212,090	\$437	\$241	88,870
SAV	Savannah, GA	\$345	\$188	205,630	\$370	\$173	114,230
HSV	Huntsville, AL	\$334	\$198	140,470	\$396	\$230	63,840
ECP	Panama City Bch, FL	\$339	\$178	138,090	\$387	\$222	36,600
CHA	Chattanooga, TN	\$316	\$36*	129,120	\$370	\$246	76,690
LEX	Lexington, KY	\$408	\$221	125,210	\$427	\$280	78,880
VPS	Destin, FL	\$317	\$186	122,210	\$377	\$154	53,470
TLH	Tallahassee, FL	\$345	\$208	106,690	\$353	\$200	42,780
GNV	Gainesville, FL	\$338	\$200	98,090	\$375	\$215	51,880
AGS	Augusta, GA	\$325	\$189	93,950	\$386	\$200	65,450
ONT	Ontario, CA	\$486	\$272	92,070	\$371	\$159	40,270
SMF	Sacramento, CA	\$475	\$288	84,560	\$386	\$190	38,130
ROC	Rochester, NY	\$371	\$218	80,220	\$403	\$195	77,040
HPN	White Plains, NY	\$333	\$157	75,380	\$469	\$287	48,960
MYR	Myrtle Beach, SC	\$318	\$163	72,380	\$347	\$202	124,180
PVD	Providence, RI	\$394	\$211	70,680	\$395	\$174	135,660
ALB	Albany, NY	\$445	\$264	70,170	\$394	\$177	86,930
DAY	Dayton, OH	\$367	\$196	67,270	\$410	\$218	51,130
SNA	Orange County, CA	\$479	\$298	67,000	\$466	\$274	30,970
BTR	Baton Rouge, LA	\$401	\$228	66,400	\$414	\$209	34,250
TUL	Tulsa, OK	\$395	\$248	64,460	\$353	\$141	50,900
XNA	Fayetteville, AR	\$428	\$268	62,790	\$419	\$222	72,580
ILM	Wilmington, NC	\$374	\$234	57,590	\$395	\$254	155,510
MSN	Madison, WI	\$418	\$276	54,210	\$412	\$224	61,600
DSM	Des Moines, IA	\$424	\$260	53,730	\$363	\$176	54,910
ROA	Roanoke, VA	\$388	\$249	48,360	\$377	\$271	74,870
CHO	Charlottesville, VA	\$390	\$218	46,160	\$381	\$254	75,170
MDT	Harrisburg, PA	\$414	\$249	43,870	\$362	\$225	105,940
PWM	Portland, ME	\$446	\$284	43,330	\$373	\$163	58,160

SHV	Shreveport, LA	\$420	\$235	41,560	\$416	\$207	33,480
GEG	Spokane, WA	\$467	\$293	39,600	\$371	\$262	7,420
SGF	Springfield, MO	\$406	\$283	38,990	\$436	\$240	47,150
FWA	Fort Wayne, IN	\$393	\$246	37,910	\$400	\$177	29,450
SBN	South Bend, IN	\$423	\$247	36,700	\$392	\$209	45,430
ABE	Allentown, PA	\$362	\$175	33,980	\$363	\$166	44,290
CID	Cedar Rapids, IA	\$371	\$257	26,240	\$424	\$297	44,210
BTV	Burlington, VT	\$416	\$293	26,180	\$409	\$234	26,880
MLI	Moline, IL	\$359	\$228	23,240	\$359	\$186	27,360
ATW	Appleton, WI	\$413	\$279	21,140	\$350	\$171	13,590